

MEAN SUBTREE ORDER AND CATERPILLAR MAXIMIZERS: AN ALL-ORDER PROOF CANDIDATE

ABSTRACT. We present a computer-assisted candidate proof of the assertion that every tree maximizing mean subtree order at a fixed order is a caterpillar. The argument selects a terminal arm of minimum rooted subtree count and propagates quantitative host bounds to every suffix of that arm. A packet-concentration comparison removes internal leaf packets, reducing the remaining configuration to a broom of unrestricted length. Exact polynomial certificates then establish the proposed broom-straightening comparison, with a separate cherry boundary argument. The complete mathematical derivation and the cherry certificates are included, together with reproducible Python code and machine-readable certificates. Fresh computations enumerate all 81,134 unlabeled trees of orders four through seventeen, check the prescribed improvements on all 64,497 non-caterpillars in that range, and generate the accompanying figures. These finite computations support implementation checking; they are not a proof of the unbounded theorem. The manuscript remains a proof candidate pending independent mathematical verification, and no complete Lean formalization is claimed.

INTRODUCTION AND MAIN STATEMENT

Jamison introduced mean subtree order and subtree density in [1, 2]. For a tree T of order n , write $\mu(T)$ for the average order of its nonempty connected induced vertex subsets and $\text{Den}(T) = \mu(T)/n$. Maximizing these two quantities is equivalent at fixed n . The path has mean $(n+2)/3$ and minimizes the mean; stars have density tending to $1/2$, whereas suitable double-brooms have density tending to one [1]. Jamison’s caterpillar conjecture asks whether some caterpillar attains the maximum at each order; this formulation is recorded as Conjecture 1 on the REGS 2011 problem page [4].

Main theorem (candidate). *For every integer $n \geq 1$, every tree on n vertices maximizing the mean order of its nonempty connected induced vertex subsets is a caterpillar. Consequently, for every n , a caterpillar attains the maximum mean subtree order among all n -vertex trees.*

The theorem is the conclusion proposed by the argument and the supplied exact certificates. Its stronger “every maximizer” formulation is distinguished from the original existence conjecture. Completeness of the written argument, reproducibility of the symbolic calculations, and mathematical correctness are separate issues. Independent confirmation of the full argument is still pending.

Related work. Mol and Oellermann [7] developed the Gluing Lemma and obtained structural restrictions on maximizing trees; their Theorem 3.9 gives the leaf-neighbor restriction used below, and their Section 2 reports computations through order 24. Cambie, Wagner, and Wang [8] proved that the maximum mean is $n - 2 \log_2 n + O(1)$. The present candidate does not invoke this asymptotic estimate or a published finite-order verification. The rooted first-moment bound and local–global comparison are due to Jamison [1, Lemma 3.2(c) and Theorem 3.9]; further local–global results appear in [5]. The inequalities $N \geq 3t - s(s+1)$ and $\mu(T) \leq (s+2)/3$, which arose in earlier exploratory notes, are already [9, Lemma 3.1 and Corollary 4.3] and are not claimed as new here.

Vince and Wang [3] studied density bounds in the class without degree-two vertices. Ralaivaosaona and Wagner [6] studied subtree-order distributions, a topic distinct from the present extremal mean question. The edge-contraction conjecture on the REGS page was resolved by Ruoyu Wang [10],

Date: 20 September 2026. Working manuscript for mathematical review.

Key words and phrases. Mean subtree order, caterpillar, extremal tree, rooted subtree, computer-assisted proof, polynomial certificate.

following [9]; that theorem concerns contraction, not the caterpillar conjecture. No result about coefficient unimodality is asserted here.

Organization and verification scope. Section 1 fixes the counting conventions and recalls elementary bounds. Section 2 selects the terminal arm and controls its retained hosts; Section 3 concentrates its last internal leaf packet. Section 4 gives the unrestricted broom comparison and its exact certificate specification, and Section 5 assembles the candidate maximality contradiction. Section 6 reports fresh computations and distinguishes them from the mathematical proof obligations. Appendix A records provenance, Appendix B contains the complete cherry argument, and Appendix C supplies additional computational plots. The named lemmas and mathematical equation tags are retained from the source manuscript to make cross-checking straightforward.

1. DEFINITIONS AND ELEMENTARY FACTS

All trees are finite, nonempty, and simple. A subtree means a nonempty vertex subset inducing a connected graph. For a tree X , let

$$N_X = \#\{\text{subtrees of } X\}, \quad R_X = \sum_{S \text{ subtree of } X} |S|, \quad \mu(X) = R_X/N_X.$$

For a specified root, let s_X, t_X be the corresponding count and order sum restricted to subtrees containing that root.

The skeleton $K(T)$ is the subgraph of the original tree induced by its non-leaf vertices. A caterpillar is a tree whose skeleton is a path, allowing an empty path or a singleton. The word $C(c_1, \dots, c_h)$ denotes a specified path of h support vertices, with c_i pendant leaves at support i , rooted at the first support. Its specified path need not equal the intrinsic skeleton of the component considered separately.

1.1. Rooted joining identities. The subtree-counting framework originates in Jamison's work [1, 2]. The edge-joining identities used here are obtained directly from the following partition, so no external counting result is invoked without proof.

Joining disjoint rooted trees A, B by an edge and retaining the root of A gives

$$\begin{aligned} s_{A+B} &= s_A(1 + s_B), & t_{A+B} &= t_A(1 + s_B) + s_A t_B, \\ N_{A+B} &= N_A + N_B + s_A s_B, & R_{A+B} &= R_A + R_B + t_A s_B + s_A t_B. \end{aligned} \quad (1.1)$$

Indeed, a connected set either lies in one component or meets both roots. Thus the formulas include all subtrees, not just the root-containing ones.

Deleting the root into components X_i gives

$$s = \prod_i (1 + s_i), \quad \frac{t}{s} = 1 + \sum_i \frac{t_i}{1 + s_i}, \quad N = s + \sum_i N_i, \quad R = t + \sum_i R_i. \quad (1.2)$$

For a fixed source tree T , always put $q = \mu(T)$. The fixed-source score of a same-order competitor Z is

$$\Delta(Z) = R_Z - qN_Z. \quad (1.3)$$

It is positive exactly when $\mu(Z) > \mu(T)$. Equivalently, its sign is the sign of the integer $R_Z N_T - R_T N_Z$.

1.2. Rooted bounds. The next first-moment inequality is Jamison's Lemma 3.2(c), and the following local-global mean comparison is his Theorem 3.9 [1]. Their proofs are included for self-containment; the statements are not claimed as new.

For every rooted tree,

$$2t \leq s(s + 1). \quad (1.4)$$

Induction in (1.2) gives $t_i/(1 + s_i) \leq s_i/2$, while $\prod_i (1 + s_i) \geq 1 + \sum_i s_i$, proving the assertion.

Also,

$$\mu(X) \leq t_X/s_X. \quad (1.5)$$

To prove this, write $L = t/s$ and $L_i = t_i/s_i$. By (1.4),

$$L - L_i = 1 - \frac{t_i}{s_i(1 + s_i)} + \sum_{j \neq i} \frac{t_j}{1 + s_j} \geq \frac{1}{2}.$$

Induction bounds each component's global mean by its rooted mean. Formula (1.2) expresses the original global mean as a weighted average of L and those component global means. This proves (1.5).

We shall use the lower bound

$$s \geq 5 \implies \frac{t}{s} \geq \frac{5}{2}. \quad (1.6)$$

Here $t \geq 2s - 1$, since only the singleton root-containing subtree has order one. If the root has at least three children, each contributes at least $1/2$ in (1.2). With two children and $s \geq 5$, one child has count at least two and contributes at least one, while the other contributes at least $1/2$. With one child, its count is $s - 1 \geq 4$. Counts at least five give a contribution at least $(2s_i - 1)/(s_i + 1) \geq 3/2$. A rooted tree of count four is either an endpoint-rooted four-vertex path or a two-leaf star rooted at its center; their rooted order sums are ten and eight. The remaining contribution is therefore at least $8/5$. These cases prove (1.6).

1.3. The leaf-neighbor restriction. This structural restriction is established in Mol and Oellermann [7, Theorem 3.9]. We give the direct score argument below rather than use that theorem as a black box.

A maximizing tree of order $n \geq 4$ has $q > 2$. For $n = 4$, the star has mean $23/11$. For $n \geq 5$, the path has mean $(n + 2)/3 > 2$, obtained by counting its intervals.

Suppose a leaf x has degree-two neighbor u , with other neighbor w . Let $H = T - \{u, x\}$, rooted at w , with $s_H = S, t_H = SM$. Move x from u to w , retaining wu . Direct use of (1.1) gives score

$$S(M + 1 - q) + q - 2 > 0, \quad (1.7)$$

because $M + 1$ is the original tree's rooted mean at w and is at least q by (1.5). This contradicts maximality.

Therefore a maximizing tree has no leaf adjacent to a degree-two vertex. In particular, every leaf of its skeleton supports at least two original leaves.

1.4. The cherry base case. We use the following all-order statement:

Cherry lemma. *In a maximizing tree of order at least four, a support with exactly two leaf neighbors and one other neighbor cannot have its other neighbor be a skeleton branching vertex.*

Appendix B reproduces its complete proof. It uses (1.1)–(1.5), the additional inductive inequality

$$6t \leq 4N + s^2 + 5s - 4|V(X)|, \quad (1.8)$$

and six exact polynomial positivity certificates. It has no order cutoff and is independent of the large-order manuscript. Its six certificates were regenerated and checked as part of this continuation.

2. MINIMUM-COUNT TERMINAL ARMS AND THEIR HOST BOUNDS

A **terminal arm** of a non-caterpillar tree is a whole component of $T - w$, where w is a branching vertex of $K(T)$, such that its vertices in $K(T)$ form a path from a neighbor of w to a leaf of $K(T)$, containing no further skeleton branching vertex. The root of the arm is the neighbor of w .

Every non-caterpillar has a terminal arm: root its skeleton at any branching vertex and choose a branching vertex farthest from that root; its outward components are terminal paths. If the root is the only branching vertex, every skeleton component at that root is such a path.

Choose a terminal arm P minimizing its rooted count, and put

$$\rho = s_P. \quad (2.1)$$

Let w be its attachment vertex.

Lemma H (Count dominance and propagation into the arm). *Choose two other non-leaf components B, C of $T - w$, and retain w and every remaining component in A . Put*

$$p = s_A, \quad x = s_B, \quad y = s_C.$$

Then

$$x, y \geq \rho. \quad (2.2)$$

Moreover, cut P immediately before any of its support vertices, obtaining a suffix of rooted count b . Let H be the whole retained tree on the other side of that edge, rooted at the suffix's attachment vertex. Write $S = s_H, M = t_H/s_H$. Then

$$\boxed{S \geq (b+1)^2, \quad \frac{M}{S} \leq \frac{1}{b+1}.} \quad (2.3)$$

Proof. Every other non-leaf component at w either is itself a terminal arm, or contains one: in the latter case choose a skeleton branching vertex farthest from w within the component and one of its outward terminal arms.

Let Q be such an arm inside B . Every subtree of Q containing its root can be extended by the fixed path from the root of B to that root. This gives an injection into root-containing subtrees of B . Thus $s_B \geq s_Q \geq \rho$. The same argument gives $s_C \geq \rho$, proving (2.2).

First delete all of P . The retained tree H_0 , rooted at w , has

$$S_0 = p(1+x)(1+y) \geq (\rho+1)^2.$$

By (1.4), its rooted mean satisfies

$$M_0 \leq \frac{p+x+y+1}{2}.$$

Consequently,

$$\frac{M_0}{S_0} \leq \frac{1}{2} \left(\frac{1}{x+1} + \frac{1}{y+1} \right) \leq \frac{1}{\rho+1}. \quad (2.4)$$

The first inequality uses $p+x+y+1 \leq p(x+y+2)$.

Now grow the retained prefix of P one support vertex at a time, treating the newly added support as root and including its original leaf packet. If the packet has size $a \geq 0$, put $r = 2^a$ and $e = 1 + a/2$. A retained rooted pair (S, M) changes to

$$S' = r(1+S), \quad M' = e + \frac{SM}{1+S}.$$

Since $e/r \leq 1$,

$$\frac{M'}{S'} = \frac{e}{r(1+S)} + \frac{SM}{r(1+S)^2} \leq \frac{1}{S} + \frac{M}{S}. \quad (2.5)$$

Also $S' \geq S + 1$.

Suppose d support vertices precede the suffix of count b . The recurrence for the original arm, read from its suffix toward its root, gives

$$\rho \geq b + d. \quad (2.6)$$

Each step adds at least one to the rooted count, regardless of its leaf packet.

All retained counts during the growth are at least S_0 . By (2.4)–(2.6),

$$\frac{M}{S} \leq \frac{1}{\rho+1} + \frac{d}{(\rho+1)^2} \leq \frac{2\rho-b+1}{(\rho+1)^2} \leq \frac{1}{b+1}.$$

The last inequality follows from the exact identity

$$(\rho + 1)^2 - (2\rho - b + 1)(b + 1) = (\rho - b)^2 \geq 0. \quad (2.7)$$

Finally $S \geq S_0 \geq (\rho + 1)^2 \geq (b + 1)^2$. This proves (2.3). $\square$

The ratio in (2.3) is $M/S = t_H/s_H^2$, not merely the rooted mean. The lemma controls the actual retained host at every cut along the selected arm; it does not replace that host by a formal tuple.

3. CONCENTRATING THE LAST INTERNAL PACKET

For integers $\ell \geq 1, a \geq 1, k \geq 2$, define

$$D_{\ell;a,k} = C(a, \underbrace{0, \dots, 0}_{\ell-1}, k).$$

There are $\ell + 1$ support vertices. Write

$$r = 2^a, \quad z = 2^k, \quad b = r(\ell + z) = s_{D_{\ell;a,k}}.$$

Lemma P (Packet concentration). *Attach $D_{\ell;a,k}$ to an arbitrary rooted host H with rooted count S and rooted mean M , obtaining T . Suppose*

$$S \geq b^2, \quad M/S \leq 1/b. \quad (3.1)$$

Then either shortening the final support into its predecessor or moving all a leaves from the first support to the final support strictly increases the global mean.

Both operations preserve the complete vertex set.

Proof of Lemma P.. Put $q = \mu(T)$, $d = q - M$, and

$$\chi_k = \frac{kz}{2(z-1)}, \quad \eta_a = \frac{ar}{2(r-1)}, \quad D_0 = r(S+1) + \ell - 2 > 0.$$

Define

$$\varepsilon = \frac{(r + \ell - 2)M + (\ell - 2)(\ell - 1 + a)/2}{D_0}. \quad (3.2)$$

The shortening comparison. The exact shortening score is

$$\Delta_{\text{short}} = (z - 1)D_0 \left(\ell + \frac{a}{2} + \chi_k - \varepsilon - d \right). \quad (3.3)$$

Here shortening makes the terminal support into an additional leaf at its predecessor. At $\ell = 1$, the replacement word is $C(a + k + 1)$.

For a direct count, let $U = C(a, 0, \dots, 0)$ be the ℓ -support prefix. Its full-prefix count and conditional mean are r and $\ell + a/2$. Its rooted count and total order at the **last** support are

$$g = r + \ell - 1, \quad j = r(\ell + a/2) + \ell(\ell - 1)/2.$$

The generating-polynomial difference for shortening is

$$((1 + u)^k - 1) [F_H(u)u^\ell(1 + u)^a + G_U(u) - u].$$

Indeed, intervals ending at the last two supports are replaced by one interval, and the last support becomes a new singleton leaf. Evaluation and differentiation at $u = 1$, followed by subtraction of q times the count increment, give (3.3). The identity

$$(g - 1)(M + \ell + a/2) - j + 1 = (r + \ell - 2)M + (\ell - 2)(\ell - 1 + a)/2$$

gives the displayed ε .

If shortening does not improve, (3.3) implies

$$d \geq d_0 := \ell + a/2 + \chi_k - \varepsilon. \quad (3.4)$$

The concentration comparison. Let G move all a leaves from the first support to the final support, producing $C(0, \dots, 0, a+k)$ on the same $\ell+1$ supports.

Its component increments, before attaching to H , are

$$\begin{aligned}\Delta s &= -\ell(r-1), & \Delta t &= -\frac{\ell}{2}[(r-1)(\ell+1) + ar], \\ \Delta N &= \ell(r-1)(z-1), \\ \Delta R &= \frac{\ell}{2}[(r-1)(z-1)(\ell+1) + ar(z-1) + kz(r-1)].\end{aligned}\tag{3.5}$$

These follow by counting spine intervals, or by substituting the two endpoint packets in the double-broom interval formula. Singleton leaf counts are unchanged.

Using (1.1), the full-tree score therefore satisfies

$$\boxed{\frac{\Delta(G)}{\ell(r-1)} = (S-z+1) \left(d - \frac{\ell+1}{2} - \eta_a \right) - (z-1)M + \frac{kz}{2}}.\tag{3.6}$$

The coefficient $S-z+1$ is positive by (3.1). We may substitute (3.4) into (3.6) in a lower-bound direction. Its parenthesized expression becomes

$$\frac{\ell-1}{2} - \frac{a}{2(r-1)} + \chi_k - \varepsilon.\tag{3.7}$$

Use $a \leq 2^a - 1 = r-1$.

Case $\ell \geq 2$. Since $r+\ell-2 \leq r\ell/2$ and $\ell-1+a \leq \ell+r-2 \leq r\ell/2$, (3.1)–(3.2) give

$$\begin{aligned}\varepsilon &\leq \frac{r+\ell-2}{r^2(\ell+z)} + \frac{(\ell-2)(\ell+r-2)}{2rS} \\ &\leq \frac{1}{2r} + \frac{\ell^2}{4S} \leq \frac{1}{2r} + \frac{1}{4r^2} \leq \frac{5}{16}.\end{aligned}\tag{3.8}$$

The first bound uses $D_0 \geq rS$ and $M \leq S/[r(\ell+z)]$.

Since $\chi_k \geq 1$, (3.7) is at least $\ell/2 - 5/16 \geq 11/16$. Also

$$(z-1)M \leq S/2, \quad kz/2 \geq z.$$

Consequently,

$$\boxed{\frac{\Delta(G)}{\ell(r-1)} \geq \frac{11}{16}(S-z+1) - \frac{S}{2} + z = \frac{3S+5z+11}{16} > 0}.\tag{3.9}$$

Case $\ell = 1$. Equation (3.2) gives

$$\varepsilon \leq \frac{(r-1)M}{rS} \leq \frac{r-1}{r^2(1+z)} \leq \frac{1}{4(1+z)} \leq \frac{1}{20}.\tag{3.10}$$

Here $(r-1)/r^2 \leq 1/4$. For every integer $k \geq 2$, $\chi_k \geq 4/3$: equality holds at $k=2$, and $k \geq 3$ gives $\chi_k > k/2 \geq 3/2$.

Thus (3.7) is at least

$$\frac{4}{3} - \frac{1}{2} - \frac{1}{20} = \frac{47}{60}.$$

Using the same bounds on $(z-1)M$ and $kz/2$,

$$\boxed{\frac{\Delta(G)}{r-1} \geq \frac{47}{60}(S-z+1) - \frac{S}{2} + z = \frac{17S+13z+47}{60} > 0}.\tag{3.11}$$

Both cases give a strict improvement whenever shortening fails. This proves Lemma P. $\square$

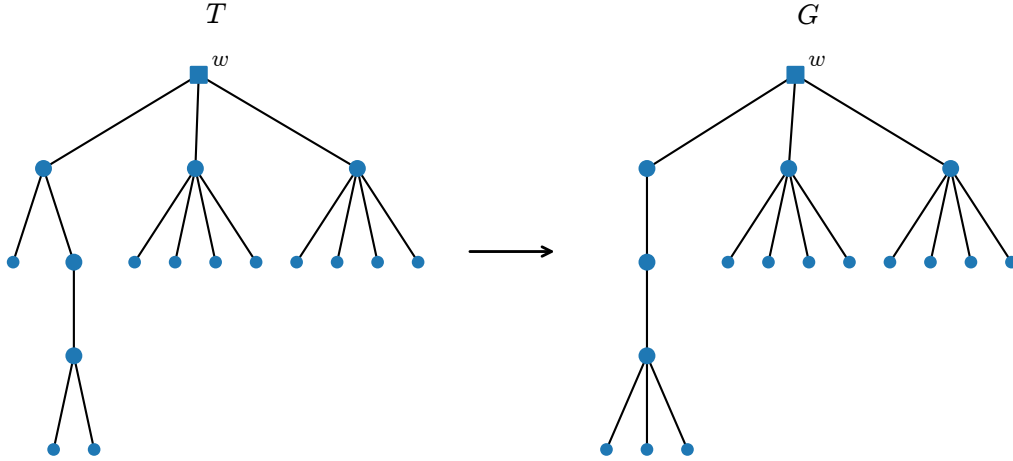


FIGURE 1. Packet concentration on an explicit 17-vertex tree. The source T has arms $C(1, 0, 2), C(4), C(4)$ at w ; the first arm becomes $C(0, 0, 3)$ in G . A square marks the retained attachment vertex, larger circles are non-leaf vertices, and smaller circles are leaves. Both drawings are generated from the exact adjacency lists used in the Python computation. The resulting graph is not itself a caterpillar; the comparison only needs a strict mean increase. The exact score is $R_G N_T - R_T N_G = 1,542,748 > 0$.

Corollary: a minimum-count terminal arm in a maximizer is a broom. Suppose the selected arm P of a maximizing tree has an internal leaf packet. Choose its **last** positive internal packet. Its suffix has the form $D_{\ell,a,k}$ above, with $a \geq 1, k \geq 2$. Lemma H gives the stronger bounds $S \geq (b+1)^2$ and $M/S \leq 1/(b+1)$, hence (3.1).

Maximality makes the shortening non-improving. Lemma P then makes the concentration strictly improving, a contradiction.

Therefore

$$P = C(\underbrace{0, \dots, 0}_{\ell}, k), \quad \ell \geq 0, \quad k \geq 2. \quad (3.12)$$

There is **no bound on ℓ** . There is also no need to preserve the minimum-count property after the concentration: the single strict improvement already contradicts maximality.

4. BROOMS OF ARBITRARY LENGTH

Write $B_{\ell,k} = C(0^\ell, k)$, with $\ell + 1$ supports. Its rooted count is

$$s_{B_{\ell,k}} = \ell + 2^k.$$

Lemma B (Count-dominated broom straightening). *Let A, B, C be arbitrary disjoint nonempty rooted trees, with root w in A . Attach B, C and $B_{\ell,k}$ at w , forming T . Suppose*

$$s_B, s_C \geq \ell + 2^k. \quad (4.1)$$

Assume either

$$\ell \geq 1, k \geq 2, \quad \text{or} \quad \ell = 0, k \geq 3.$$

Then one of these same-order changes strictly increases the global mean:

1. Shorten the final support of the broom into its predecessor when $\ell \geq 1$; when $\ell = 0$, fold its k leaves onto w , retaining the star center as one more leaf at w .

2. *Retain all $\ell + 1$ supports as a bare path from w to the root of B , moving B 's attachment from w to the terminal support. Distribute the original k terminal leaves as evenly as possible between the roots of B, C . Permit both choices of the moved branch and both balanced allocations.*

The proof of this lemma is a finite algebraic certificate for an unbounded parameter range. We specify the entire calculation next.

4.1. Variables and the exact shortening bound. Put

$$\begin{aligned} p &= s_A, & \alpha &= t_A/p = 1 + c/2, & 0 &\leq c \leq p-1, \\ x &= s_B, & y &= s_C, & U &= \frac{t_B}{x(x+1)}, & V &= \frac{t_C}{y(y+1)}. \end{aligned}$$

The variable c is a real host-moment parameter, not a leaf count. Equations (1.4) and (1.6) imply

$$\frac{5}{2(x+1)} \leq U \leq \frac{1}{2}, \quad \frac{5}{2(y+1)} \leq V \leq \frac{1}{2}. \quad (4.2)$$

The rooted counts are at least five in the nonstar case, and at least eight in the star case, so (1.6) applies.

Let

$$\begin{aligned} z &= 2^k, & S &= p(x+1)(y+1), & M &= 1 + c/2 + xU + yV, & d &= q - M, \\ D &= S + \ell - 1 > 0, & \chi_k &= \frac{kz}{2(z-1)}. \end{aligned}$$

The symbol D here is a denominator, not $n - q$.

The exact shortening/folding score is $(z-1)D(d_0 - d)$, where

$$d_0 = \ell + \chi_k - \frac{(\ell-1)M + (\ell-1)(\ell-2)/2}{S + \ell - 1}. \quad (4.3)$$

For $\ell \geq 1$, this is the same interval count used in (3.3), specialized to an all-zero prefix. For $\ell = 0$, it becomes

$$d_0 = \chi_k + \frac{M-1}{S-1},$$

which follows directly from the fold polynomial $(F_H(u) - u)((1+u)^k - 1)$.

Thus failure of the first move gives $d \geq d_0$.

4.2. Complete insertion counts. Allocate b terminal leaves to the moved branch B , and $k-b$ to C . Put

$$L = 2^b, \quad J = 2^{k-b}, \quad LJ = z, \quad h = \ell + 1.$$

For this paragraph write $t_B = x(x+1)U$ and $t_C = y(y+1)V$.

The old broom and new path-plus- B component have rooted moments

$$\begin{aligned} s_P &= \ell + z, & t_P &= \ell(\ell+1)/2 + z(h + k/2), \\ s_Q &= h + Lx, & t_Q &= h(h+1)/2 + L[t_B + x(h + b/2)]. \end{aligned} \quad (4.4)$$

The old and new root-containing moments at w are

$$\begin{aligned} s_0 &= S(1 + s_P), & t_0 &= s_0M + St_P, \\ s_1 &= p(1 + s_Q)(1 + Jy), \\ t_1 &= s_1(1 + c/2) + p(1 + Jy)t_Q + p(1 + s_Q)J[t_C + (k-b)y/2]. \end{aligned} \quad (4.5)$$

The **full global increments**, including subtrees that avoid w , are

$$\Delta N = s_1 - s_0 + [(h+1)L - 1]x + (J-1)y + h(1-z), \quad (4.6)$$

$$\boxed{\begin{aligned} \Delta R = & t_1 - t_0 + [(h+1)L - 1]t_B + (J-1)t_C \\ & + \frac{Lx(h+1)(h+b)}{2} + \frac{Jy(k-b)}{2} \\ & + \frac{h(h+1)(1-z)}{2} - \frac{zkh}{2}. \end{aligned}} \quad (4.7)$$

For an independent derivation of the terms avoiding w , adding b leaves at the root of B changes its global moments by

$$N_B \mapsto N_B + (L-1)x + b, \quad R_B \mapsto R_B + (L-1)t_B + bLx/2 + b.$$

Appending a bare path of h new vertices adds $h(h+1)/2$ path-only subtrees and hLx mixed subtrees; their order sums are $h(h+1)(h+2)/6$ and $hL(t_B + bx/2) + Lxh(h+1)/2$. The original broom has

$$N_P = k + \ell(\ell+1)/2 + zh, \quad R_P = k + \ell(\ell+1)(\ell+2)/6 + zh(h+1+k)/2.$$

Subtracting these expressions, and adding the change inside C , gives exactly (4.6)–(4.7). The verifier checks the same identities by a separate symbolic joining calculation.

The fixed-source score of this insertion is

$$\Sigma_b = \Delta R - (M+d)\Delta N.$$

Let $\mathcal{I}$ be its arithmetic average over both insertion sides and the allocations $b \in \{\lfloor k/2 \rfloor, \lceil k/2 \rceil\}$, counting an even allocation once. It is an average of scores using the **same original** q , not an average of different means.

If $\mathcal{I} > 0$, at least one actual insertion strictly improves.

4.3. The finite polynomial assertion. The certificate verifies

$$\boxed{\partial_d \mathcal{I} > 0, \quad \mathcal{I}(U, V, d_0) > 0} \quad (4.8)$$

throughout the parameter range of Lemma B and the rectangle (4.2).

Here is a complete specification of the polynomials and their proof. The expressions (4.4)–(4.7), their stated average, and (4.3) uniquely define everything that follows.

Put

$$\mathcal{H} = 8(z-1)D\mathcal{I}(U, V, d_0). \quad (4.9)$$

It is affine in U, V . Define

$$\begin{aligned} U_0 &= \frac{5}{2(x+1)}, \quad U_1 = \frac{1}{2}, \quad V_0 = \frac{5}{2(y+1)}, \quad V_1 = \frac{1}{2}, \\ Q_{ij} &= (x+1)(y+1)\mathcal{H}(U_i, V_j). \end{aligned} \quad (4.10)$$

Every cleared factor is strictly positive. By affinity, positivity at the four corners proves positivity on the whole rectangle. Symmetry makes Q_{01} the variable-swapped version of Q_{10} , so only three corners require separate certificates.

Nonstars with $k \geq 8$: unbounded in both k and ℓ . Write

$$k = 2g + e, \quad e \in \{0, 1\}, \quad v = 2^g, \quad g \geq 4.$$

The balanced powers are $(L, J) = (v, v)$ for $e = 0$, and $(v, 2v), (2v, v)$ for $e = 1$. Also $z = (1+e)v^2$.

Introduce independent nonnegative variables

$$P = p - 1 - c, \quad C = c, \quad H = \ell - 1, \quad X = x - \ell - z, \quad Y = y - \ell - z. \quad (4.11)$$

For either parity and each of the three corners, expansion of Q_{ij} gives 576 nonzero monomials in P, C, H, X, Y . Grouping by P, C, X, Y leaves **96 polynomials in H** :

$$Q_{ij} = \sum_{\nu} P^{\nu_1} C^{\nu_2} X^{\nu_3} Y^{\nu_4} \sum_j f_{\nu j}(g, v) H^j. \quad (4.12)$$

Every $f_{\nu j}$ is affine in g .

Put $W = v - 16 \geq 0$. For every integer $g \geq 4$,

$$\boxed{4 \leq g \leq v/4 = (W + 16)/4.} \quad (4.13)$$

Indeed $2^4 = 4 \cdot 4$, and doubling $2^g \geq 4g$ preserves the inequality at $g + 1$.

Write a coefficient after $v = W + 16$ as

$$f = f_0(W) + g[f_+(W) + f_-(W)],$$

where f_+ has nonnegative coefficients and f_- has nonpositive coefficients, obtained by splitting the coefficient of g by monomial sign. Then

$$\boxed{f \geq B(W) := f_0(W) + 4f_+(W) + \frac{W + 16}{4}f_-(W).} \quad (4.14)$$

The exact difference is

$$(g - 4)f_+(W) + \left(g - \frac{W + 16}{4}\right)f_-(W) \geq 0.$$

This identity, including its direction, is checked for each coefficient.

For each grouped polynomial in (4.12), the lower polynomial

$$\sum_j B_j(W)H^j$$

has one of the following certificates:

- Every coefficient is nonnegative and $B_0(0) > 0$; or
- All B_j , except possibly B_1 , have nonnegative coefficients, and an explicitly recorded positive rational λ gives

$$\begin{aligned} \sum_j B_j H^j &= B_2[H - \lambda(W + 16)]^2 \\ &\quad + [B_1 + 2\lambda(W + 16)B_2]H \\ &\quad + [B_0 - \lambda^2(W + 16)^2 B_2] + \sum_{j \geq 3} B_j H^j. \end{aligned} \quad (4.15)$$

Both bracketed remainder polynomials have nonnegative coefficients, and the second has a strictly positive constant. Thus the expression is strictly positive for every $H, W \geq 0$.

Every one of the 576 grouped certificates is included in the data files. This counts two parities, three symmetry-distinct corners, and 96 groups. Their 3,456 coefficient lower bounds, exact expressions, square centers, and remainder polynomials are all recorded. No group or coefficient is omitted.

A representative certificate is short enough to display. In the even, upper–upper corner, the coefficient of $P^2 X^3 Y^3$ is

$$\begin{aligned} G(H, g, v) &= 4(v^2 - 1)H^2 \\ &\quad + (8gv^2 - 4v^3 + 12v^2 + 4v - 12)H \\ &\quad + 16gv^2 + 8v^4 - 16v^3 + 16v - 8. \end{aligned}$$

Using $g \geq 4$,

$$\begin{aligned} G(H, g, v) &\geq 4(v^2 - 1)(H - v/2)^2 + (44v^2 - 12)H \\ &\quad + 7v^4 - 16v^3 + 65v^2 + 16v - 8 > 0. \end{aligned} \quad (4.16)$$

The final polynomial is positive for $v \geq 16$: $7v^4 - 16v^3 = v^3(7v - 16) > 0$, and the remaining terms are positive. The other recorded certificates use exactly the general form (4.15).

For transparency, the square-center ledger is:

Parity	Corner	Certificates
Even	lower–lower	96 coefficient-positive groups
Even	upper–lower	$\lambda = 1/4$: 48; $3/8$: 32; $1/2$: 16
Even	upper–upper	$\lambda = 1/2$: 48; $3/4$: 32; 1: 13; $9/8$: 3
Odd	lower–lower	96 coefficient-positive groups
Odd	upper–lower	$\lambda = 3/8$: 48; $5/8$: 32; $3/4$: 16
Odd	upper–upper	$\lambda = 3/4$: 48; $9/8$: 32; $3/2$: 13; $13/8$: 3

The data associates each center with its precise monomial index. The centers are not numerical fits: substitution gives exact identities with rational coefficients, and every remainder coefficient is checked.

The same substitution proves $\partial_d \mathcal{I} > 0$ by nonnegative-coefficient expansion, with a strictly positive constant term. This has no dependence on U, V .

Equations (4.12)–(4.15) establish (4.8) for every $k \geq 8$ and every $\ell \geq 1$. There is no upper bound on either parameter.

Nonstars with $2 \leq k \leq 7$. Substitute the exact powers $z = 2^k, L = 2^b, J = 2^{k-b}$. Keep ℓ , the other branch counts, and the host unbounded. After (4.11), group each corner polynomial by P, C, X, Y as above. Each of its 96 coefficient polynomials has the form

$$a_0 + a_1 H + a_2 H^2 + \cdots.$$

Either all coefficients are nonnegative with $a_0 > 0$, or only a_1 is negative, $a_2 > 0$, and

$$4a_0 a_2 - a_1^2 > 0.$$

In the latter case the exact decomposition is

$$a_2 \left(H + \frac{a_1}{2a_2} \right)^2 + \frac{4a_0 a_2 - a_1^2}{4a_2} + \sum_{j \geq 3} a_j H^j > 0. \quad (4.17)$$

Terminal packet k	Coefficient-positive groups	Quadratic-square groups	Total groups
2	288	0	288
3	288	0	288
4	285	3	288
5	273	15	288
6	261	27	288
7	225	63	288

All 1,728 boundary groups, including their exact coefficients and the 108 positive discriminants, are recorded. The derivative $\partial_d \mathcal{I}$ is also coefficient-positive in every boundary case. This proves (4.8) at the remaining packets.

Stars: $\ell = 0, k \geq 3$. Set $\ell = 0$ in (4.3)–(4.10) and use

$$P = p - 1 - c, \quad C = c, \quad X = x - z, \quad Y = y - z.$$

At $k = 3$, each of the three corner numerators and the derivative has nonnegative coefficients and a positive constant. This is a fixed packet calculation with arbitrary host and branch counts.

For even $k = 2g$ and odd $k = 2g + 1, g \geq 2$, set $v = 2^g, W = v - 4$, and use

$$2 \leq g \leq v/2.$$

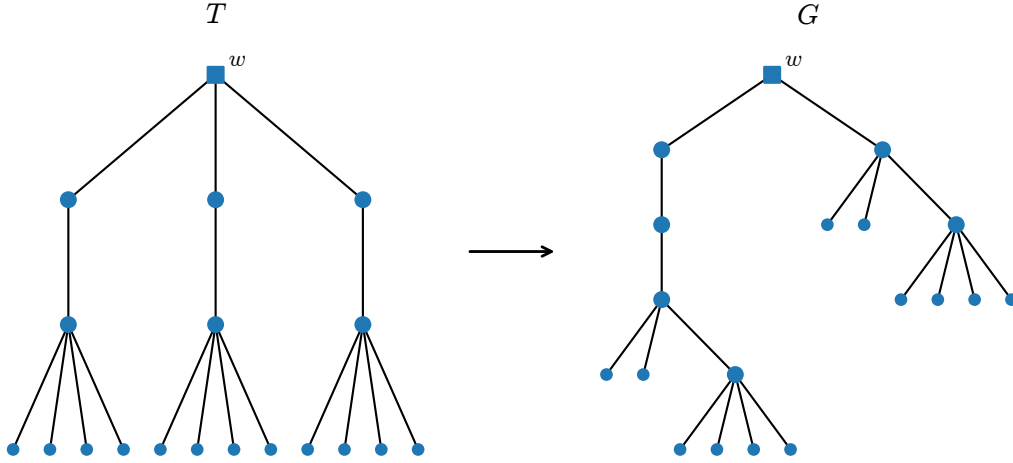


FIGURE 2. Balanced insertion on the 19-vertex tree with three arms $C(0, 4)$ at w . Two terminal leaves of the selected arm are reattached at each of the other branch roots, and its two support vertices remain in the connecting path. The resulting caterpillar is $C(4, 2, 0, 0, 0, 2, 4)$. Vertex sizes and the square have the same meaning as in Figure 1. The exact comparison score is $6,755,700 > 0$.

Apply the coefficient-sign splitting (4.14), replacing 4 and $(W + 16)/4$ by 2 and $(W + 4)/2$. All 576 corner-coefficient lower polynomials have nonnegative coefficients; their base coefficients have positive constants. The derivative certificates have the same property. No square in a path-length variable is needed because the arm has one support.

This proves (4.8) for every star packet $k \geq 3$.

4.4. Conclusion of Lemma B. If shortening or folding improves, the first alternative holds. Otherwise $d \geq d_0$. The positive derivative in (4.8) and the corner certificates imply

$$\mathcal{I}(U, V, d) \geq \mathcal{I}(U, V, d_0) > 0.$$

At least one of the actual balanced insertions therefore has strictly positive fixed-source score. All components remain connected by single attachment edges, and all leaf changes are reattachments; each competitor is a tree on exactly the original vertices.

This proves Lemma B, with the finite algebraic part supplied by (4.4)–(4.17) and the accompanying exact certificates. $\square$

5. COMPLETION OF THE CANDIDATE ARGUMENT

We now prove the theorem.

Every tree of order at most three is a caterpillar. Fix $n \geq 4$ and let T be an arbitrary maximizer of mean subtree order among n -vertex trees. A maximum exists because there are finitely many labeled trees of fixed order.

Suppose T is not a caterpillar. By Section 1.3, no leaf has a degree-two neighbor. Select a terminal arm P of minimum rooted subtree count as in Section 2. Its terminal leaf packet is at least two.

If P has a positive internal leaf packet, its last such packet supplies the suffix in Lemma P. Lemma H supplies the required host bounds. Maximality makes shortening non-improving, so concentration strictly improves, a contradiction. Consequently $P = B_{\ell, k}$ for some $\ell \geq 0, k \geq 2$.

If $\ell = 0, k = 2$, P is a terminal cherry at a skeleton branching vertex, contradicting the cherry lemma.

In every other case, choose two other non-leaf components B, C at its attachment vertex and retain everything else in A . Lemma H gives

$$s_B, s_C \geq s_P = \ell + 2^k.$$

All hypotheses of Lemma B hold. That lemma gives a same-order tree with strictly greater global mean, contradicting maximality.

Every case for $\ell \geq 0, k \geq 2$ has been included. Therefore T is a caterpillar. Since the chosen maximizer was arbitrary and every contradiction was strict, **every** maximizer is a caterpillar. The existence statement follows immediately. $\square$

Where the infinite ranges are handled. There is no induction that merely advances the path-length cutoff one step. Arbitrary internal decoration is removed by Lemma P, whose inequalities hold for every suffix length. Arbitrary broom length is the free nonnegative variable $H = \ell - 1$ in the coefficient and square certificates. Arbitrary terminal packet size is covered by the two parities, the inequalities $2^g \geq 4g$, and the finite boundary packets. Host and other branch sizes remain independent unbounded variables throughout.

6. COMPUTATIONAL EVIDENCE AND REPRODUCIBILITY

6.1. Exact arithmetic and the role of computation. The computation has two distinct purposes. First, the supplied SymPy verifier [11] reconstructs the polynomial identities and sign certificates used in Section 4 and Appendix B. Second, finite graph computations test the implementations and illustrate the transformations. Only the first purpose is part of the proposed all-order computer-assisted argument; no finite graph test, plot, or sampling endpoint establishes the infinite theorem.

For the fresh execution accompanying this edition, the runtime used Python 3.13.5, SymPy 1.14.0, NetworkX 3.6.1 and Matplotlib 3.10.8. The full supplied verifier was rerun, exited successfully, and generated the results in `verification/checks/`. All five regenerated certificate data files were byte-for-byte identical to their supplied counterparts. This is a reproducibility check on the same program, not an independent verification of its mathematical specification. Table 1 summarizes the exact checks.

The verifier also reproduces 30,000 seeded configurations and 1,440 systematic broom comparisons, with test endpoints up to 1,000 bare support vertices and 1,024 terminal leaves. These are supporting tests only. A test call can verify multiple properties of a configuration; the counts are not estimates of a probability of correctness. Exact integer or rational arithmetic is used for all sign decisions.

TABLE 1. Freshly reproduced algebraic checks. Counts describe certificate objects, not numbers of tree orders covered. Unbounded domains are justified by the inequalities in the mathematical argument, not by the object counts.

Certificate family	Count
Grouped unbounded broom certificates	576
Coefficient lower-bound identities for those groups	3,456
Small-packet broom groups, with unrestricted length	1,728
Unbounded star corner coefficients	576
Fixed three-leaf-star corner coefficients	288
Cherry vertex certificates	6
Packet-concentration and host identities recorded by the verifier	5

TABLE 2. Exact finite extrema from the fresh enumeration. The non-caterpillar column counts all inputs in that class, not only extremal examples. A dash denotes an empty non-caterpillar class. These data do not extend the published order-24 verification; they reproduce a smaller range while checking the present transformations.

n	All trees	Non-caterpillars	M_n	B_n
4	2	0	23/11	—
5	3	0	13/5	—
6	6	0	117/37	—
7	11	1	131/35	10/3
8	23	3	583/135	4
9	47	11	779/159	554/119
10	106	34	205/37	959/180
11	235	99	2025/329	404/67
12	551	279	4236/619	3313/493
13	1,301	773	2513/335	943/127
14	3,159	2,103	10243/1247	3958/489
15	7,741	5,661	2383/267	326/37
16	19,320	15,160	23713/2453	4559/478
17	48,629	40,373	26596/2555	30421/2961

Floating-point numbers enter only when the verified finite data are converted to coordinates for plotting.

6.2. Fresh enumeration and independent recounting. The new driver `analysis/run_experiments.py` enumerates each unlabeled tree of orders $4 \leq n \leq 17$ using NetworkX [12]. Subtree moments are recomputed from the adjacency sets by a separate implementation of the root-deletion recurrence. For every non-caterpillar, the unchanged supplied transformation routine returns an explicit same-order competitor. The new driver checks connectivity, acyclicity, order preservation, and the comparison score by recounting that competitor. The two implementations use the same mathematical counting identity, so this check is implementation-level cross-checking rather than independence of the underlying mathematics.

Let $\mathcal{T}_n$ be the enumerated tree class and $\mathcal{C}_n$ its caterpillar subclass. The finite statistics are

$$M_n = \max_{T \in \mathcal{T}_n} \mu(T), \quad B_n = \max_{T \in \mathcal{T}_n \setminus \mathcal{C}_n} \mu(T), \quad \delta_n = M_n - B_n \quad (n \geq 7). \quad (6.1)$$

The class defining B_n is empty for the enumerated orders four through six; no zero or fabricated value is assigned there. All comparisons in (6.1) use cross-multiplied integers. The fresh run confirms that every observed maximizer is a caterpillar in this finite range. The exact extrema appear in Table 2; their densities and gaps are shown in Figures 3 and 4.

For a non-caterpillar input T , write $G(T)$ for the competitor chosen by the supplied routine. The routine maximizes the positive integer comparison score among its listed candidates; it does not necessarily maximize the resulting mean. The observed gain is

$$g(T) = \mu(G(T)) - \mu(T) = \frac{R_{G(T)}N_T - R_T N_{G(T)}}{N_T N_{G(T)}} > 0. \quad (6.2)$$

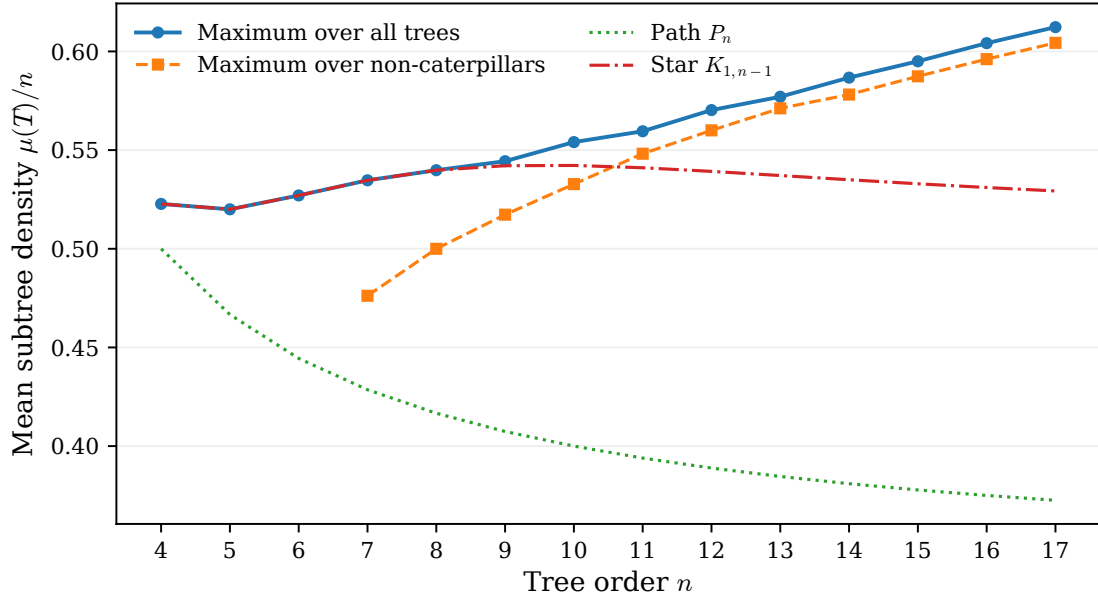


FIGURE 3. Exact finite extremal densities for $4 \leq n \leq 17$, compared with paths and stars. All extrema are found by enumeration; the path and star curves use their exact subtree-count formulas. Non-caterpillar values begin at $n = 7$. Connecting segments guide the eye between integer orders; they do not represent an interpolation theorem or an asymptotic bound.

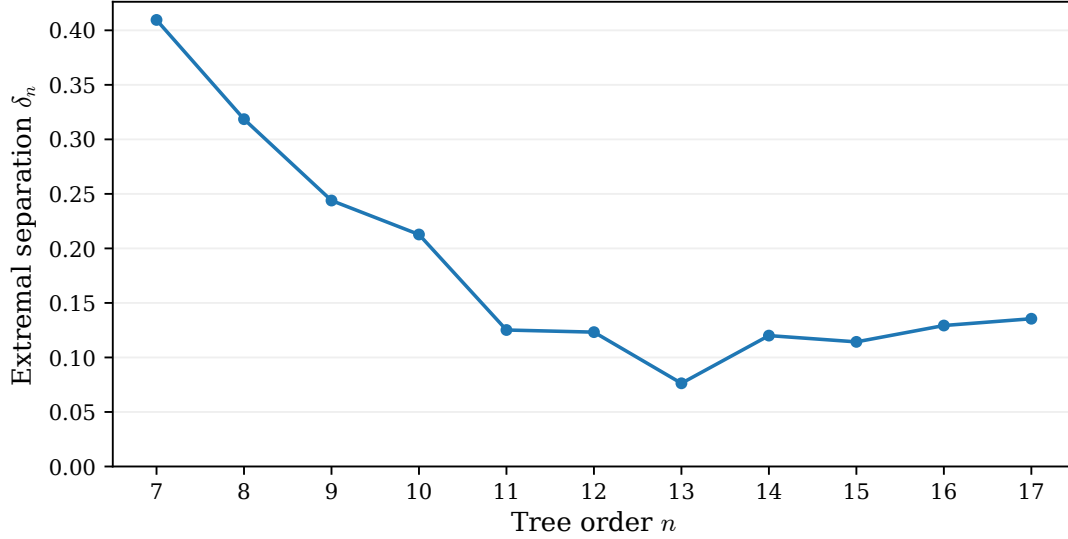


FIGURE 4. The finite extremal separation $\delta_n = M_n - B_n$ from (6.1). Its positivity for $7 \leq n \leq 17$ was decided by exact rational arithmetic. The variation is a finite observation and gives no uniform lower bound for untested orders.

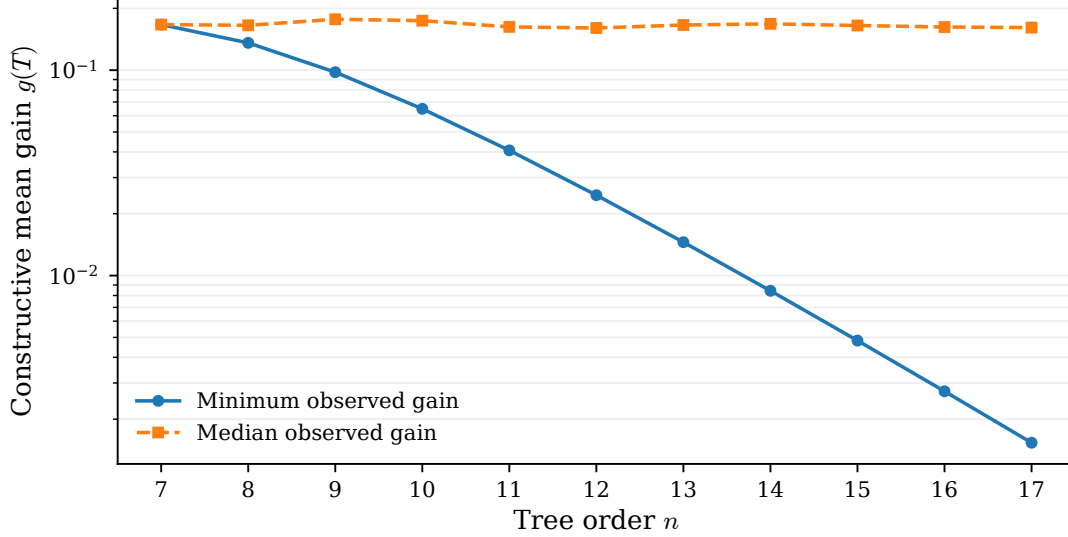


FIGURE 5. Minimum and median observed gain $g(T)$ over all non-caterpillar inputs of each enumerated order. The vertical axis is logarithmic. Every plotted value is derived from exact source and competitor counts, with no tolerance used in positivity decisions. A decreasing observed minimum is not evidence of a limiting rate.

TABLE 3. Classification of the 64,497 non-caterpillar inputs by the first applicable case of the supplied routine. The cases are disjoint in the implementation; the counts are not intrinsic invariants of the input trees.

Case used	Inputs
Leaf adjacent to a degree-two vertex	61,844
Minimum-count arm is a cherry	1,808
Minimum-count arm is a nonstar broom	585
Minimum-count arm is a larger star	240
Minimum-count arm has an internal packet	20
Total non-caterpillars	64,497

For each order, Figure 5 plots the minimum and median of these exact gains over the non-caterpillar inputs. In the run there were 81,134 trees, 16,637 caterpillars, and 64,497 non-caterpillars. Every non-caterpillar had a strictly improving returned competitor. At $n = 17$, the minimum observed gain was 783352/510207147, approximately 0.00153536; this is a run statistic and not an all-order estimate.

6.3. Exact examples and subtree-order distributions. The graph drawings in Figures 1 and 2 come from the actual source and competitor adjacency lists. For both comparisons, a separate bitset algorithm tests the connectivity of every nonempty vertex subset. This yields the full sequences $a_j(T)$ and $a_j(G)$, not just their means, with

$$N_T = \sum_{j=1}^n a_j(T), \quad R_T = \sum_{j=1}^n j a_j(T), \quad p_j(T) = \frac{a_j(T)}{N_T}. \quad (6.3)$$

The seventeen-vertex concentration example gives $(N_T, R_T) = (3821, 37861)$ and $(N_G, R_G) = (3249, 32597)$, with score

$$32597 \cdot 3821 - 37861 \cdot 3249 = 1542748 > 0.$$

The nineteen-vertex insertion example gives $(N_T, R_T) = (5943, 69363)$ and $(N_G, R_G) = (5218, 62038)$, with score

$$62038 \cdot 5943 - 69363 \cdot 5218 = 6755700 > 0.$$

Together the four graphs require 1,310,716 nonempty subset tests. Their normalized distributions are plotted in Appendix C. A larger mean is the comparison proved by these exact finite counts; no stochastic domination or unimodality conclusion is asserted.

6.4. Reproduction and limits of the verification. The archive accompanying this manuscript contains the original verifier, regenerated certificate files, raw graph-level data, plotting code, exact summary tables, and execution logs. From the archive root the computations can be reproduced with

```
python -m pip install -r requirements.txt
python verification/verify_jamison_all_orders.py --part all --output-dir
↪ verification/checks
python analysis/run_experiments.py --max-order 17
python analysis/plot_results.py
latexmk -pdf -interaction=nonstopmode -halt-on-error main.tex
```

The main data files are `data/experiments.json`, `data/noncaterpillar_moves.csv`, and `data/enumeration_summary.csv`. The raw comparison file identifies both graphs in graph6 format and records N_T, R_T, N_G, R_G and the exact rational gain for every tested non-caterpillar. The machine-readable certificate files list every term and square center. Hashes in the manifest establish file identity only; they are not mathematical evidence.

The principal audit obligations remain the minimum-count injection and host propagation in Lemma H, the global score in Lemma P, the insertion identities, the signs and completeness of the certificate substitutions in Lemma B, and the cherry boundary argument. The written candidate presents an argument for all these obligations, but rerunning its own verifier cannot exclude a shared specification error. The argument and its initial code were developed with language-model assistance; independent human mathematical confirmation is not claimed. The previously supplied Lean development remains partial and has not been compiler-checked, and it is not a proof of the main graph theorem. This edition adds typesetting and reproducible computational evidence, not a new claim of independent correctness or journal acceptance.

APPENDIX A. SOURCE AND DEPENDENCY LEDGER

The project-file provenance below is separate from published mathematical attribution. In particular, the rooted first-moment and local-global inequalities are credited to Jamison [1], and the leaf-neighbor restriction to Mol and Oellermann [7]. The references and historical discussion record related work without making that work an unstated premise or treating an internal project source as an independent publication.

The original supplied `jamison_caterpillar_manuscript.pdf` provides the terminology, rooted joining identities, rooted first-moment upper bound, local-versus-global mean inequality, leaf-neighbor restriction, and spine-interval counting convention. The needed elementary arguments are restated in Section 1 and the exact comparisons above.

The earlier supplement `jamison_all_order_cherry_lemma.md` supplies the cherry base case. Its mathematical argument is reproduced in Appendix B, with local numbering, and its six certificate calculations were regenerated in the fresh execution reported in Section 6.

The **minimum-rooted-count selection**, the **host-ratio propagation**, the **last-internal-packet concentration lemma**, the **count-dominated arbitrary-length broom and star certificate**, and the **all-order assembly** constitute the proposed main argument of this manuscript.

The original candidate's asymptotic theorem, benchmark construction, numerical cutoff, and strengthened logarithmic deficit are not used. Neither are the later three-leaf, all-terminal-star, or two-vertex-arm supplements. The present candidate is therefore not relying on an unverified extension of those fixed-length arguments.

APPENDIX B. COMPLETE CHERRY BASE-CASE PROOF

The following reproduces the mathematical proof from the earlier cherry supplement. Its lemma numbers are local to this appendix. Its local result is the cherry lemma used in Section 1.4, not the whole all-order theorem. The proof is included so that the main chain does not rely on an unstated small-star assertion.

B.1. Notation and elementary rooted facts. For a nonempty rooted tree X , write

$$(m, s, t, N, R) = (|V(X)|, s_X, t_X, N_X, R_X), \quad \mu(X) = R/N.$$

The quantities s, t count root-containing subtrees and sum their orders; N, R do the same for all nonempty connected induced vertex sets.

If the components after deleting the root have moments $(m_i, s_i, t_i, N_i, R_i)$, then

$$s = \prod_i (1 + s_i), \quad t = s \left(1 + \sum_i \frac{t_i}{1 + s_i} \right), \quad N = s + \sum_i N_i, \quad R = t + \sum_i R_i, \quad m = 1 + \sum_i m_i. \quad (\text{B.1})$$

These formulas partition subtrees according to whether they contain the root.

Lemma B.1 (Rooted first moment). *For every rooted tree,*

$$2t \leq s(s + 1). \quad (\text{B.2})$$

Proof. Induct on the order. For a singleton equality holds. Otherwise, induction and (B.1) give

$$\frac{t}{s} = 1 + \sum_i \frac{t_i}{1 + s_i} \leq 1 + \frac{1}{2} \sum_i s_i \leq \frac{1 + s}{2},$$

because $\prod_i (1 + s_i) \geq 1 + \sum_i s_i$. Multiply by $2s$. □

Lemma B.2 (Local mean dominates global mean). *For every choice of root,*

$$R/N \leq t/s. \quad (\text{B.3})$$

Proof. Put $L = t/s$, $L_i = t_i/s_i$. By (B.1)–(B.2),

$$L - L_i = 1 - \frac{t_i}{s_i(1 + s_i)} + \sum_{j \neq i} \frac{t_j}{1 + s_j} \geq \frac{1}{2}.$$

Induction gives $R_i/N_i \leq L_i < L$. Formula (B.1) expresses the global mean as a weighted average of L and these component means. □

B.2. A count–moment inequality retaining unrooted information.

Lemma B.3. *Every finite nonempty rooted tree satisfies*

$$\boxed{6t \leq 4N + s^2 + 5s - 4m.} \quad (\text{B.4})$$

Consequently, since $m \geq t/s$,

$$\boxed{N \geq \left(\frac{3}{2} + \frac{1}{s}\right)t - \frac{s^2 + 5s}{4}.} \quad (\text{B.5})$$

Proof. Use induction on m . A singleton gives equality in (B.4). For a root-deletion component use the notation in (B.1), and put

$$p_i = \frac{s}{1 + s_i} \geq 1, \quad A = \sum_i s_i, \quad Q = \sum_i s_i^2.$$

The inductive hypotheses give

$$4N_i - 4m_i \geq 6t_i - s_i^2 - 5s_i.$$

Substitute in the left-hand side of (B.4), rearranged:

$$\begin{aligned} 4N + s^2 + 5s - 6t - 4m &\geq s^2 + 3s - 4 - Q - 5A - 6 \sum_i (p_i - 1)t_i \\ &\geq s^2 + 3s - 4 + 2Q - 2A - 3sA. \end{aligned} \quad (\text{B.6})$$

The second inequality uses $2t_i \leq s_i(s_i + 1)$ and $p_i - 1 \geq 0$.

It remains to show

$$\mathcal{G}(p, A, Q) := p^2 + 3p + 2Q - 2A - 3pA \geq 4 \quad \text{when } p = \prod_i (1 + s_i). \quad (\text{B.7})$$

For the empty list, $p = 1, A = Q = 0$, so $\mathcal{G} = 4$. Append an integer $z \geq 1$ to the list. The change is

$$z[-3Ap + p^2z + 2p^2 - 3pz + 2z - 2].$$

Since $A \leq p - 1$, it is at least

$$z(z - 1)(p - 1)(p - 2) \geq 0.$$

Indeed, $p = 1$ before any factor has been added, and afterwards the integer $p \geq 2$. Thus (B.7) follows by induction on the number of factors. Equations (B.6)–(B.7) prove (B.4). Finally substitute $m \geq t/s$ into (B.4), rearranging to obtain (B.5). $\square$

The important distinction from a rooted-deficit estimate alone is that (B.4)–(B.5) keep N , the number of **all** subtrees. A proposed combination of rooted count and rooted mean therefore forces a minimum unrooted count as well.

B.3. Exact cherry comparisons. Let H be a tree rooted at w . Add new vertices u, ℓ_1, ℓ_2 and edges $wu, u\ell_1, u\ell_2$, obtaining T . Write $q = \mu(T)$.

The **fold** F moves ℓ_1, ℓ_2 from u to w , retaining wu . If wz is an edge of H , the **swing** C deletes wz and adds uz .

The rooted three-vertex cherry has moments

$$(s, t, N, R) = (4, 8, 6, 10).$$

Partitioning at the attachment edge gives, with $S = s_H$ and $M = t_H/S$,

$$N_F - N_T = 3(S - 1), \quad R_F - R_T = 3MS + 4S - 7. \quad (\text{B.8})$$

For the swing, let A, B be the components of $H - wz$, rooted at w, z . Put

$$a = s_A, \quad \alpha = t_A/a, \quad b = s_B, \quad \beta = t_B/b.$$

Then

$$\begin{aligned} N_C - N_T &= b(4 - a), \\ R_C - qN_C &= b[(a - 4)(q - \beta) - a\alpha + 8]. \end{aligned} \quad (\text{B.9})$$

All scores use the fixed original mean q ; positive score is exactly strict improvement.

Lemma B.4 (Sufficient cherry-improvement condition; restored factor). *If*

$$25a^2 + 48a - 120t_A \geq 0, \quad (\text{B.10})$$

then the fold or the swing strictly improves T .

Proof. Suppose the fold does not improve. Since $S = a(1 + b) > 1$ and $M = \alpha + b\beta/(1 + b)$, (B.8) implies

$$q - M \geq \frac{4}{3} + \frac{q - 7/3}{S}. \quad (\text{B.11})$$

As $M \geq 1$, rearranging proves $q \geq 7/3$. Lemma B.1 gives $\beta/(1 + b) \leq 1/2$.

Root T at z . Its local mean is

$$\beta + \frac{5a}{1 + 5a} \left(\alpha + \frac{8}{5} \right),$$

so Lemma B.2 gives

$$q - \beta - 2 < \alpha - \frac{2}{5}. \quad (\text{B.12})$$

By (B.9), the swing score divided by ab equals

$$q - M - \frac{\beta}{1 + b} - \frac{4}{a}(q - \beta - 2).$$

Using (B.11)–(B.12), this is strictly greater than

$$\frac{5}{6} - \frac{4\alpha}{a} + \frac{8}{5a} = \frac{25a^2 + 48a - 120t_A}{30a^2} \geq 0.$$

Thus the swing strictly improves. □

B.4. The three-branch lemma for every order.

Theorem B.5. *Let B, C be arbitrary nonempty rooted trees satisfying $s_B, s_C \geq 4$. Form T by attaching B, C and a three-vertex cherry to a new vertex w , and also attach $c \geq 0$ singleton leaves to w .*

Then at least one of the following strictly increases the mean subtree order while preserving the vertex set:

1. *Fold the two cherry leaves onto w .*
2. *Move the entire branch B from w to the cherry center.*
3. *Move the entire branch C from w to the cherry center.*

There is no order cutoff.

Proof for $c \geq 2$. Apply Lemma B.4 to the edge attaching B . The retained side A consists of w , its c leaves, and C . Set $y = s_C$. Then

$$a = 2^c(1 + y) \geq 5 \cdot 2^c, \quad \alpha = 1 + \frac{c}{2} + \frac{t_C}{1 + y} \leq \frac{c + 1}{2} + \frac{a}{2^{c+1}}.$$

Consequently,

$$\frac{25a^2 + 48a - 120t_A}{a} \geq (25 - 60/2^c)a - 60c - 12 \geq 125 \cdot 2^c - 60c - 312 > 0.$$

The final expression is 68 at $c = 2$, and increases thereafter. Lemma B.4 applies.

Proof for $c = 0, 1$: exact reduction to two variables. Put

$$x = s_B, \quad y = s_C, \quad U = \frac{t_B}{x(x+1)}, \quad V = \frac{t_C}{y(y+1)}.$$

By Lemma B.1,

$$x, y \geq 4, \quad 0 < U, V \leq \frac{1}{2}. \quad (\text{B.13})$$

Write

$$p = 2^c, \quad S = p(1+x)(1+y), \quad M = 1 + \frac{c}{2} + xU + yV, \quad d = q - M.$$

Suppose all three changes are non-improving. Equation (B.8) gives

$$d \geq d_0 := \frac{4}{3} + \frac{c/2 + xU + yV}{S-1}. \quad (\text{B.14})$$

The two swing inequalities from (B.9), weakened by replacing d by d_0 , are

$$G_B := [p(1+y) - 4](d_0 - U) - 4yV + 4 - 2c \leq 0, \quad (\text{B.15})$$

$$G_C := [p(1+x) - 4](d_0 - V) - 4xU + 4 - 2c \leq 0. \quad (\text{B.16})$$

Both replacement coefficients $p(1+y) - 4$ and $p(1+x) - 4$ are positive, so this weakening is valid.

The local mean of B is $(x+1)U$, and similarly for C . The subtrees of T avoiding w lie in B, C , in the cherry, or are one of the c singleton leaves. Thus

$$\begin{aligned} 0 = R_T - qN_T &= 5S(8/5 - d) + (R_B - qN_B) + (R_C - qN_C) \\ &\quad + 10 + c - (6+c)q. \end{aligned} \quad (\text{B.17})$$

Apply Lemma B.2 to B, C . The resulting upper bound on (B.17) is strictly decreasing as a function of d , with derivative

$$-[5S + N_B + N_C + 6 + c] < 0.$$

It may therefore be further bounded above by its value at $d = d_0$. Set

$$q_0 = M + d_0.$$

Notice

$$q_0 - (x+1)U = 1 + c/2 + yV + d_0 - U > 0,$$

and likewise with B, C exchanged. Now, and only now, use Lemma B.3 to replace N_B, N_C by the respective lower bounds

$$\mathcal{B}_x(U) = \left(\frac{3}{2}x + 1\right)(x+1)U - \frac{x^2 + 5x}{4}, \quad \mathcal{B}_y(V) = \left(\frac{3}{2}y + 1\right)(y+1)V - \frac{y^2 + 5y}{4}. \quad (\text{B.18})$$

This is valid even if a lower bound happens to be negative, since the coefficients just displayed are positive. The order of the preceding steps matters: monotonicity in d was used with the actual positive counts, not with possibly negative relaxed counts.

We obtain the necessary inequality

$$\boxed{0 \leq \mathcal{F}_c(U, V)}, \quad (\text{B.19})$$

where

$$\begin{aligned} \mathcal{F}_c(U, V) &= 5S(8/5 - d_0) \\ &\quad - \mathcal{B}_x(U)[q_0 - (x+1)U] - \mathcal{B}_y(V)[q_0 - (y+1)V] \\ &\quad - (6+c)q_0 + 10 + c. \end{aligned} \quad (\text{B.20})$$

We next prove that (B.19) is impossible under (B.13), (B.15), and (B.16).

Why only four vertices need checking. For fixed x, y, c , the constraints define a compact polygon $\mathcal{P}$ in $[0, 1/2]^2$. Both coefficients in each of the affine inequalities $G_B, G_C \leq 0$ are negative. No point on $U = 0$ or $V = 0$ is feasible: for example, at $U = 0, V \leq 1/2$,

$$G_C \geq [p(1+x) - 4](4/3 - 1/2) + 4 - 2c > 0.$$

The upper corner $(1/2, 1/2)$ satisfies both inequalities strictly.

The common intersection of $G_B = G_C = 0$ is $I = (U_I, V_I)$. For $c = 0$,

$$U_I = \frac{4(x+2)}{15(x+1)}, \quad V_I = \frac{4(y+2)}{15(y+1)}. \quad (\text{B.21})$$

For $c = 1$,

$$U_I = \frac{16x^2y + 4x^2 + 24xy + x + 2y - 5}{6(x+1)(6xy + x + y - 1)}, \quad (\text{B.22})$$

and V_I is obtained by exchanging x, y . These lie strictly between zero and $1/2$. In (B.22), positivity is immediate at $x, y \geq 4$, and the denominator minus twice the numerator is

$$2(x-1)(2xy - x - y - 2) > 0.$$

The right vertex is $J = (1/2, V_J)$, with

$$V_J = \frac{5xy + 17x + 5y + 9}{6(4x + 3)(y + 1)} \quad (c = 0), \quad (\text{B.23})$$

$$V_J = \frac{10xy^2 + 15xy - x + 10y^2 + 10y - 2}{6y(4x + 3)(y + 1)} \quad (c = 1). \quad (\text{B.24})$$

In each case $0 < V_J < 1/2$. Substituting this right vertex into the other constraint gives

$$G_C(J) = -\frac{x(7x-1)}{2(4x+3)} < 0 \quad (c = 0),$$

$$G_C(J) = -\frac{(x-1)(2xy-x-y-2)}{3y(4x+3)} < 0 \quad (c = 1).$$

The top vertex is obtained by exchanging x, y and U, V . These, together with I and the upper corner, are exactly the four vertices of $\mathcal{P}$.

The function $\mathcal{F}_c$ is quadratic in U, V . Its U^2, V^2 coefficients are positive and its UV coefficient is negative. To see this directly, put

$$A_x = (3x/2 + 1)(x + 1), \quad A_y = (3y/2 + 1)(y + 1), \quad d_x = x/(S - 1), \quad d_y = y/(S - 1).$$

The three coefficients are

$$A_x(1 - d_x) > 0, \quad A_y(1 - d_y) > 0, \quad -[A_x(y + d_y) + A_y(x + d_x)] < 0.$$

There is no interior local maximum, since restriction to a horizontal line is strictly convex. Every boundary edge is horizontal, vertical, or has negative slope. Restriction to any such edge is convex as well: if its direction is (a, b) with $ab \leq 0$, all three quadratic contributions to its second derivative are nonnegative. Therefore the maximum of $\mathcal{F}_c$ occurs at a vertex.

The vertex certificates. Appendix B.6 supplies exact polynomials proving

$$\mathcal{F}_c(I) < 0, \quad \mathcal{F}_c(J) < 0, \quad \mathcal{F}_c(1/2, 1/2) < 0 \quad (c = 0, 1). \quad (\text{B.25})$$

The top vertex follows by symmetry. These are polynomial inequalities for **every real** $x, y \geq 4$, not a finite sample of counts. Consequently $\mathcal{F}_c < 0$ on the entire polygon, contradicting (B.19). At least one move strictly improves, completing the proof of Theorem B.5. $\square$

B.5. Excluding terminal cherries at skeleton branch vertices.

Corollary B.6. *Let T be an order- n mean-subtree-order maximizer, with $n \geq 4$. Suppose u has precisely two leaf neighbors and one other neighbor w . Then*

$$\deg_{K(T)}(w) \leq 2. \quad (\text{B.26})$$

Proof. First recall the elementary leaf-neighbor restriction from manuscript Lemma 4.1. A maximizer of order $n \geq 4$ has $q > 2$: the four-vertex star proves this at $n = 4$, and the path mean $(n+2)/3$ proves it for $n \geq 5$. If a leaf z has degree-two neighbor v , let a, b be rooted count and sum of $T - \{v, z\}$ at the remaining neighbor of v . Moving z to that remaining neighbor has score

$$a \left(\frac{b}{a} + 1 - q \right) + q - 2 > 0.$$

Here $b/a + 1$ is the local mean of the original tree at that remaining neighbor and is at least q by Lemma B.2. This contradicts maximality.

Consequently, every component of $T - w$ rooted at a non-leaf neighbor of w has rooted count at least four. Indeed, its order is at most its rooted count (grow a rooted connected set through each order). A nontrivial component of rooted count at most three must be a two-vertex path or a three-vertex path rooted at an endpoint, either of which would exhibit a leaf with a degree-two neighbor in T .

Suppose $\deg_{K(T)}(w) \geq 3$. Besides the cherry at u , there are at least two other non-leaf components.

If there are exactly two, Theorem B.5 applies, including every possible number of singleton leaves at w , and produces a strict improvement.

If there are at least three, select one of them as B in Lemma B.4. The retained A at w has at least two child components of rooted counts at least four. Put $a = s_A$, $\alpha = t_A/a$. Its child factors $1 + s_i$ are all at least two, with at least two at least five. Thus $a \geq 25$ and

$$\sum_i s_i \leq a/5 + 3.$$

For completeness, retain one factor $f \geq 5$ and combine the rest into a product $g \geq 5$. The sum of the original factors minus one is at most $(f-1) + (g-1)$, and

$$f + g - 2 \leq fg/5 + 3$$

follows from $(f-5)(g-5) \geq 0$.

By Lemma B.1,

$$\alpha \leq 1 + \frac{1}{2} \sum_i s_i \leq a/10 + 5/2.$$

Hence

$$25a^2 + 48a - 120t_A \geq a(13a - 252) > 0,$$

since $a \geq 25$. Lemma B.4 again supplies a strict improvement. Both cases contradict maximality, proving (B.26). $\square$

B.6. Exact positivity certificates. Set $X = x - 4 \geq 0$, $Y = y - 4 \geq 0$. At each indicated vertex, $-\mathcal{F}_c$ is the shifted numerator below divided by the displayed denominator, which is positive for $x, y \geq 4$. For the upper and right vertices, the omitted top vertex follows by exchanging the two branches.

Central leaf count 0, middle vertex.

$$-\mathcal{F}_c = \frac{P(X, Y)}{900}.$$

A sum-of-squares decomposition with nonnegative remaining coefficients is

$$\begin{aligned} P(X, Y) = & 356 (X - Y)^2 + [36X^2Y + 103X^2 + 36XY^2 \\ & + 279X + 103Y^2 + 279Y \\ & + 13464]. \end{aligned}$$

Its strictly positive constant is 13464, proving strict positivity throughout the quadrant.

Central leaf count 0, upper vertex.

$$-\mathcal{F}_c = \frac{P(X, Y)}{12xy + 12x + 12y}.$$

All coefficients of the following expanded numerator are nonnegative:

$$\begin{aligned} P(X, Y) = & 3X^3Y^2 + 38X^3Y + 118X^3 \\ & + 3X^2Y^3 + 62X^2Y^2 + 571X^2Y \\ & + 1705X^2 + 38XY^3 + 571XY^2 \\ & + 3984XY + 10454X + 118Y^3 \\ & + 1705Y^2 + 10454Y + 24336. \end{aligned}$$

Its strictly positive constant is 24336, proving strict positivity throughout the quadrant.

Central leaf count 0, right vertex.

$$-\mathcal{F}_c = \frac{P(X, Y)}{12(4x + 3)^2}.$$

A sum-of-squares decomposition with nonnegative remaining coefficients is

$$\begin{aligned} P(X, Y) = & 458 (X - Y)^2 + [20X^4Y + 304X^4 + 6X^3Y^2 \\ & + 149X^3Y + 5288X^3 + 114X^2Y^2 \\ & + 33X^2Y + 37296X^2 + 705XY^2 \\ & + 131700X + 964Y^2 + 705Y \\ & + 185730]. \end{aligned}$$

Its strictly positive constant is 185730, proving strict positivity throughout the quadrant.

Central leaf count 1, middle vertex.

$$-\mathcal{F}_c = \frac{P(X, Y)}{72(6xy + x + y - 1)^2}.$$

All coefficients of the following expanded numerator are nonnegative:

$$\begin{aligned} P(X, Y) = & 480X^4Y^3 + 8724X^4Y^2 + 48048X^4Y \\ & + 83463X^4 + 480X^3Y^4 + 9040X^3Y^3 \\ & + 116574X^3Y^2 + 645519X^3Y + 1174982X^3 \\ & + 8724X^2Y^4 + 116574X^2Y^3 + 972832X^2Y^2 \\ & + 4594312X^2Y + 8055637X^2 + 48048XY^4 \\ & + 645519XY^3 + 4594312XY^2 + 18628422XY \\ & + 30064868X + 83463Y^4 + 1174982Y^3 \\ & + 8055637Y^2 + 30064868Y + 45250668. \end{aligned}$$

Its strictly positive constant is 45250668, proving strict positivity throughout the quadrant.

Central leaf count 1, upper vertex.

$$-\mathcal{F}_c = \frac{P(X, Y)}{24xy + 24x + 24y + 12}.$$

All coefficients of the following expanded numerator are nonnegative:

$$\begin{aligned} P(X, Y) = & 6X^3Y^2 + 82X^3Y + 263X^3 \\ & + 6X^2Y^3 + 92X^2Y^2 + 912X^2Y \\ & + 3035X^2 + 82XY^3 + 912XY^2 \\ & + 5844XY + 16729X + 263Y^3 \\ & + 3035Y^2 + 16729Y + 40570. \end{aligned}$$

Its strictly positive constant is 40570, proving strict positivity throughout the quadrant.

Central leaf count 1, right vertex.

$$-\mathcal{F}_c = \frac{P(X, Y)}{72y^2(4x + 3)^2}.$$

All coefficients of the following expanded numerator are nonnegative:

$$\begin{aligned} P(X, Y) = & 240X^4Y^3 + 4344X^4Y^2 + 23232X^4Y \\ & + 38784X^4 + 216X^3Y^4 + 4704X^3Y^3 \\ & + 65526X^3Y^2 + 353688X^3Y + 612096X^3 \\ & + 4050X^2Y^4 + 55649X^2Y^3 + 519128X^2Y^2 \\ & + 2518229X^2Y + 4291602X^2 + 24267XY^4 \\ & + 321386XY^3 + 2418429XY^2 + 10130572XY \\ & + 16183752X + 47058Y^4 + 653276Y^3 \\ & + 4587808Y^2 + 17386724Y + 25904712. \end{aligned}$$

Its strictly positive constant is 25904712, proving strict positivity throughout the quadrant.

APPENDIX C. ADDITIONAL COMPUTATIONAL DIAGNOSTICS

The plots in this appendix are generated by `analysis/plot_results.py` from the fresh exact data described in Section 6. They serve as diagnostics and illustrations, not as additional premises in the all-order argument. Each figure’s underlying integer counts and rational probabilities are included in the data archive.

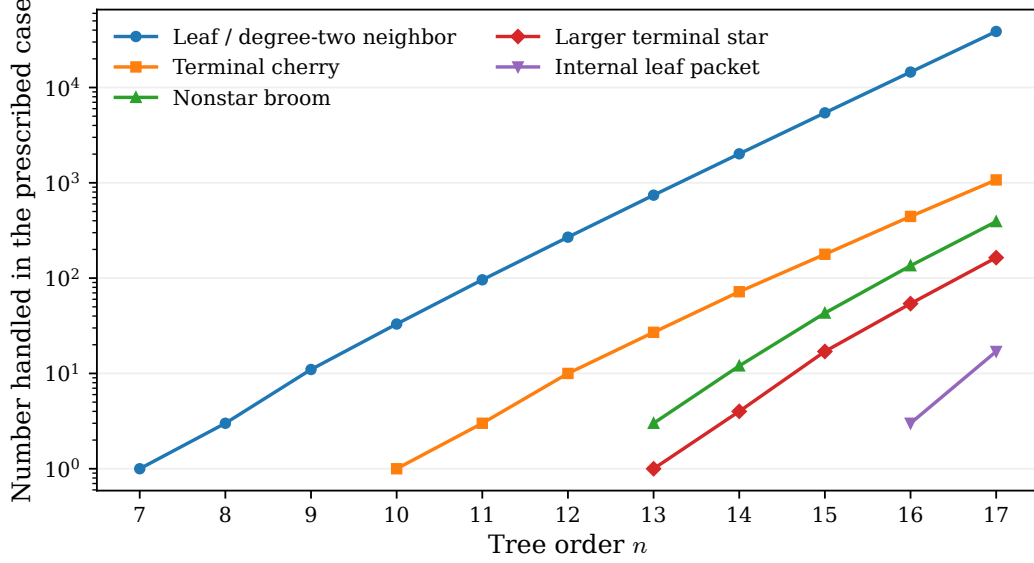


FIGURE 6. Counts by order for the prescribed first-applicable cases of the transformation routine. Zero counts are omitted on the logarithmic scale. The curves partition the tested non-caterpillars according to the algorithm’s case order, and therefore are not a census of all configurations that occur inside those trees. Aggregate counts appear in Table 3.

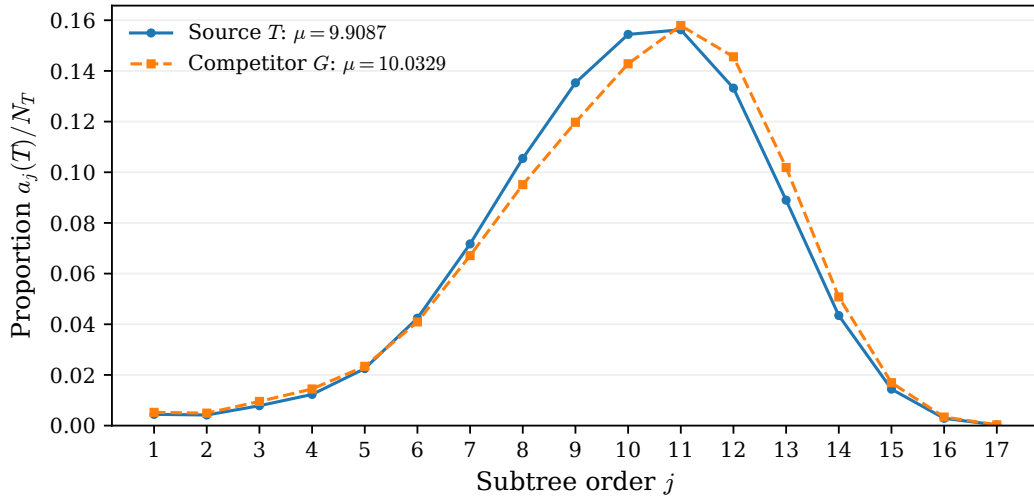


FIGURE 7. Subtree-order proportions for the seventeen-vertex packet-concentration example in Figure 1. The values come from exhaustive connectivity tests on all nonempty vertex subsets of each graph. The exact mean increase is $1542748/(3821 \cdot 3249) > 0$. Lines connect integer subtree orders only; the diagram is not a density estimation procedure.

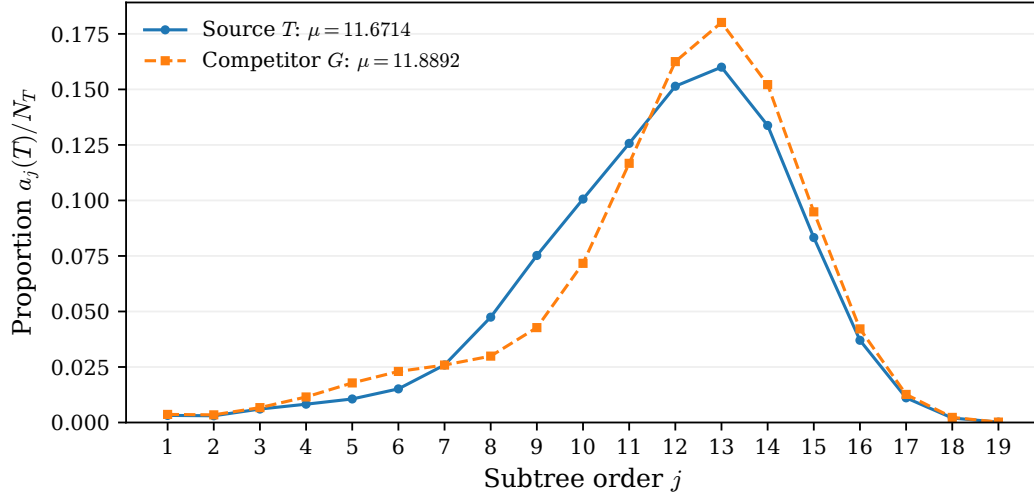


FIGURE 8. Subtree-order proportions for the nineteen-vertex balanced insertion in Figure 2. The exact mean increase is $6755700/(5943 \cdot 5218) > 0$, with $N_T = 5943$ and $N_G = 5218$. The two curves have different normalizing subtree counts; only the stated mean comparison is inferred.

TABLE 4. Exact subtree-order counts behind Figures 7 and 8, obtained by exhaustive vertex-subset connectivity tests. The seventeen-vertex columns have no entries at orders 18 and 19. Totals and first moments recover the displayed comparisons without rounding.

j	17-vertex concentration		19-vertex insertion	
	$a_j(T)$	$a_j(G)$	$a_j(T)$	$a_j(G)$
1	17	17	19	19
2	16	16	18	18
3	30	31	36	35
4	47	47	49	60
5	86	76	63	93
6	162	133	90	120
7	274	218	154	135
8	403	309	282	156
9	517	389	447	223
10	590	464	598	374
11	597	513	747	609
12	509	473	900	848
13	340	331	951	940
14	166	165	795	794
15	55	55	495	495
16	11	11	220	220
17	1	1	66	66
18	—	—	12	12
19	—	—	1	1
N	3821	3249	5943	5218
R	37861	32597	69363	62038

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